

Aman Javed

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Portfolio Link(amanjaved.vercel.app)

Summary

Motivated Electrical Engineering student with hands-on experience in embedded systems, PCB design, IoT, microcontrollers, and hardware-software integration. Skilled in schematic design, PCB layout, firmware development (C/C++), debugging, and testing of electronic systems. Experienced in prototyping and implementing reliable engineering solutions for real-world applications.

Education

Aligarh Muslim University, Bachelor of Technology in Electrical Engineering Aug 2023 – Present
University Polytechnic, Aligarh Muslim University, Diploma in Electrical Engineering Jul 2020 – Jun 2023

- CPI: 9.83/10.0 - Awarded **Gold Medal** by the university

Experience

PCB Design Intern, Taipei Medical University – Taipei, Taiwan Mar 2026 – Apr 2026

- Completed a research internship at the Department of Biomedical Informatics in PCB design for biomedical applications, under the direct supervision of Prof. Dr. Shabbir Syed-Abdul
- Engaged in PCB design tasks within the department's research projects, from schematic capture through to layout
- Signed letter of recommendation available on request; project specifics are covered by a non-disclosure agreement

Coordinator, AMURoboclub – Aligarh May 2024 – Present

- Coordinated technical activities, workshops, and competitions related to **robotics, embedded systems, and PCB design**
- Assisted in organizing inter-university technical events and hands-on training sessions

EPS Team, SS AMU SAT – Aligarh, Uttar Pradesh, India Dec 2023 – Nov 2025

- Joined as an **EPS Team Member** and contributed to the design, analysis, and validation of the satellite **Electrical Power System (EPS)**
- Promoted to **EPS Designer L2** based on technical contributions and subsystem responsibility

Trainee, 3.0 MW Solar Power Plant – Aligarh Muslim University Aug 2023 – Jun 2024

- Completed short-term industrial training focused on **solar power generation**, system operation, and basic maintenance practices

Technical Projects

ESP32 Fault-Tolerant Homing Tracker (Homer) 2026

- Built a dual-failover IoT tracker on **ESP32-S3**: GPS to BLE peer positioning, and Wi-Fi to GSM (**SIM800L GPRS**) uplink, for continuous location reporting
- Implemented **BLE** advertising/scanning state machine with RSSI-based range estimation, plus **HTTP-over-AT** upload with per-transport failure recovery
- Conducted a **SIM800L** hardware study (voltage, 2A current bursts, decoupling) to address the module's known brownout failure modes

PCB Design 2024

- Designed and developed custom PCBs using **KiCad** for embedded systems and motor control applications
- Key projects include a **battery charger circuit** and a **motor controller board**, from schematic to layout

- Emphasized **signal integrity**, **robust power regulation**, and **compact, manufacturable layouts** suitable for real-world deployment

Raspberry Pi Projects

2023-2024

- Customized Linux-based OS images for embedded systems using automated build tools
- Deployed and configured **SatNOGS**-based ground station software on Raspberry Pi platforms
- Debugged and resolved OS and distribution compatibility issues in Debian-based Linux environments

Micromouse Robot Design

2024

- Selected and integrated: **TB6612FNG** motor driver, **Pololu Micro Metal Gear Motors**, and **5× VL53L0X** ToF sensors
- Built a **P-Channel MOSFET-based** step-down converter (IRF9540N + zener diode design)
- Programmed efficient navigation using wall-following logic and sensor feedback

IoT Power Monitor & Relay Control

2024

- Implemented real-time power monitoring and control systems using **ESP32-C6**
- Enabled wireless data communication and remote control using **Wi-Fi**

Maixduino Camera Streaming

2024

- Enabled video streaming on Maixduino with no display using Wi-Fi/Bluetooth
- Experimented with low-latency wireless protocols for embedded vision systems

Skills

Embedded Systems: ESP32, Raspberry Pi, Maixduino

PCB Design: KiCAD, Autodesk Eagle

Programming: C/C++, Python

IoT & Web: HTML, Flask, WebSockets

Electronics: Buck converters, sensor interfacing, power management

Tools: Git, VS Code, Excel, MATLAB

Languages: English (Fluent), Hindi (Native), Urdu (Native)

Workshops & Training

- **Conducted PCB Design Workshop for SS AMUSAT** – Delivered focused training on **KiCad-based** PCB design for satellite subsystems
- **Conducted Hands-on PCB Design Workshop (AMU Roboclub)** – Led practical **KiCad** sessions for robotics-oriented PCB design

Competitions & Achievements

- Served as Co-organizer for the **Fastest Line Follower** competition at **VERCERA**, organized by **AMU Roboclub**
- Served as Co-organizer for the **National Space Day Quiz Competition**

Certifications

- MATLAB Onramp
- Excel Mastery: Basic to Advance
- Industrial Training in Web Development
- PCB Design with KiCAD

References

Available on Request